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Investigating leadership discussion on social media: A NLP and machine learning perspective

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ABSTRACT

The explosion of digital world has created a unique avenue for modern day social scientists and researchers that has allowed them to radically assess the relationships between people, as well as act as prosumers to these relationships as well. Compared to traditional media, social media is data and information rich, which has helped exploring human behavior at an unprecedented scale. A particular area that is being explored is leadership and leader usage of social media. This study tries to investigate a user's social media behavior surrounding leaders and leadership as a whole. It uses 54,326 tweets extracted using preferred keywords to perform analysis to evaluate the semantic orientation of these tweets and predict the causative drivers through generalized linear modeling for user's behavior. The findings suggest that Twitter users have a general positive sentiment towards leadership, and there are four distinct topics that explain this relationship. This was further validated through predictive analytics, and it showed that there a statistically significant impact of sentiment polarity on the user's behavior. This study proposes a scheme for evaluating the causative factors of Twitter user's behavior which provides us with insights into the use of social media for leaders and leadership.

1. Introduction

The way of communication between people has gone through a sweeping change in the last quarter-century, especially in the previous decade and a half. These changes have been diametric in determining the intensified as well as transparent discourse. In the same vein, they have created decentralized, open, accessible, and unbarred communication across multiple channels in organizations and society. Due to this widespread use of online mediums of communication, a bilateral and bi-directional flow of information and communication has been created using internet-based communication platforms known as Social Media (Mayfield, 2008; Wang et al., 2017). Organizations have increasingly used these interactions to create a sublet of ideas, information, and knowledge to stay up-to-date with the industry, as well as the people they are dealing with (Gilani et al., 2020; Shankar & Tewari, 2023).

With the use of pervasive social media and social media platforms, forms of organizing, such as places of online discourse, exchange of ideas or information, or even town square discussions, have radically emerged through the internet communities or online communities (Benkler, 2006; Johnson et al., 2015; Preece, 2000). This creates a unique opportunity for the leaders and people in positions of power to access their credibility, decision-making capability, timeliness of their decisions, necessary actions, as well as their abilities to make tough, necessary, and important changes (Bruns et al., 2013; Goel & Sharma, 2021; Pair et al., 2021; Shankar et al., 2017).

Effective large collectibles such as large online communities bring together people who are essentially strangers to each other across nationalities, ethnicities, demographics, psychographics, or boundaries – both geographical as well as organizational (Johnson et al., 2015). These networks have an ample amount of real-time information that pertains to real events in every person's life, which are heavily characterized by the shaping of users' viewpoints using Opinion Leaders (Agarwal & Toshniwal, 2020; Katz & Lazarsfeld, 2017), and sometimes real leaders. As Goleman (2004) states that a leader is someone who can reach and inspire a huge number of people through their opinions – and thus, it becomes important that leaders – political, organizational, or otherwise – use the internet's medium to promulgate their points of view.

Despite this substantial growth in understanding – that social media plays an inherently integral role in building a person's viewpoints; there has been scarce growth in the knowledge and examination of the points of differences and similarities between an online transmission of information versus a traditional collaboration in a leader's context. This is

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especially prevalent in understanding the kinds of leadership that are pervasive in the general online discourse (Johnson et al., 2015; Shankar & Tewari, 2021). This extends to the lack of understanding of how people discuss leaders dealing with emergencies – such as a pandemic or natural disaster (Agarwal & Toshniwal, 2020; Goel & Sharma, 2021), determining and dealing with gender, racial or ethnic discrimination (Pair et al., 2021; Tabbush & Gaona, 2017; Grant, 2017), legislative politics (Ebanks, Yan, Alvarez, Das, & Sinclair, 2021; Rajput, 2014), religious and spiritual examination (Bonilla & Rosa, 2015; Udupa, 2016), or organizational sustenance, team management and leader-member exchange (Faraj et al., 2014; Gilani et al., 2020; Shankar and Tewari, 2021, 2023).

And then there is the increasing challenge of misinformation and disinformation. Public figures, experts, and scientists worldwide have extensively accredited the rise of misinformation and disinformation to the rise of communities in online spheres that are highly engaging, with fluid boundaries, emergent roles, authorities with experts, high turnover, overwhelming charisma, and singular or few leader positions providing a communal experience to the outcasts that are usually ostracized by the online public spheres (Faraj et al., 2011; Goel & Sharma, 2021; Johnson et al., 2015; Ren et al., 2007). In a societal context, much of the discourse amongst the people of differing opinions is derived from, in parts, their leader's messaging or narratives (Bode & Vraga, 2015); and in an organizational context, job-level factors are, in part, derived by the decisions of business leaders (Bertrand & Schoar, 2003; Choudhury et al., 2020).

The importance of good leaders becomes distinctively imperative when good ideas become scarce and harder to find – in an organizational setting, medical setting, scientific setting, political setting, to urban and rural setting (Azoulay et al., 2010; Caines et al., 2017; Jones & Olken, 2005; Shankar et al., 2021), especially when the audience is comparatively unknown and the tasks at hand are becoming increasingly difficult and complex (Dessein et al., 2016). There are multiple platforms where opinion leaders can gain credibility and disseminate information, generate accelerated marketing solutions, develop a criminal network, spread medical and non-medical rumors, form a Q&A community, or even try and sway elections (Agarwal & Toshniwal, 2020; Bovet et al., 2018; Bovet & Makse, 2019; Leskovec et al., 2007; Lü et al., 2011; Zhao et al., 2018).

Previous works performed in the area have unanimously found that a robust and dynamic systems design and management can help in reducing the impact of the cascading effects this circumstance creates (Perc et al., 2019). However, many of the previous works on understanding today's leaders through the prism of online discourse have dealt with a lot of manual work in identifying the patterns through the categories of extraction and classification (Imran et al., 2016; Gilani et al., 2020). Applications of human behavior in decision-making can be found in AI, psychology, and economics. Traditional approaches concentrate on outcomes, but new studies indicate that language has a significant impact on decisions. Though generative AI and sentiment analysis show the limits of current techniques (e.g., over-inflating the importance of leadership, confounding frequency of interaction with intensity), they also highlight the vitality of language in molding public engagement and behavior (Capraro et al., 2024). Using social media analytics to capture the sentiments of the users and leaders regarding their understanding of leaders and leadership may present more consistent and accurate engagement results (Malik et al., 2016) as compared to using traditional methods to understand leadership.

This study thus tries to examine the leadership of online discourse. It uses the microblogging leader – Twitter's (Twitter, 2022) tweets related to leaders and leadership and tries to analyze beyond just political and organizational leadership, and what that constitutes in an online platform. Given the lack of physical presence of the leaders, this interaction is bound to be drastically different from the more traditional form of interactions that leaders have with others, given the prevalence of "text-based asynchronous exchanges" (Johnson et al., 2015) over more familiar synchronous exchanges. This platform was chosen above its competitors as it is considered to be the fastest-growing social media platform (Sharma et al., 2017) and has been used by a plethora of users for legions of reasons, ranging from casual banters, to mental health issues, and expressing opinions that are of ideological or political nature about a brand/organization/celebrity (Chae, 2015).

The motivation behind this study was to understand and create a framework that can be used to better understand the leadership discussions in the social network, and how these discussions are driving the essential decision-making of the users engaged in those discussions. This study can be used to build upon the understanding of leaders and leadership in the online community. Since leadership has now been deemed emergent (Collier & Kraut, 2012), none of the definitions of leadership stands true and consistent with today's environment, especially when leaders are borderless, hierarchy-less, and governance-less (Goel & Sharma, 2021; Johnson et al., 2015; Shankar & Tewari, 2021; Von Krogh et al., 2012). The collected tweets cover the discussions and interactions amongst individuals, professionals, leaders – business and otherwise, and organizations covering the concepts of leader and leadership. The findings have tried to answer these questions:

(1) What are the patterns between the information dissemination related to leader and leadership tweets?
(2) What are the dominant topics, concepts, and contents on Twitter regarding leadership?
(3) What are the sentimental inclinations of online users towards leaders and leadership?
(4) To what magnitude is the user's social media behavior swayed by Twitter's supposed leaders' discussion regarding leaders and leadership?

Based on the results of the study, it tries to propose a potential method that leaders, as well as organizations, may use to disseminate more aligned information to transform the polarized opinions of online users. Even though this study is not the first in its domain, it has tried to provide more comprehensive findings to understand online leaders and online leadership.

This study is, thus, divided into the following sections: Section 2 presents the current literature about Twitter and its use in interdisciplinary domains; Section 3 presents the data collection and preprocessing; Section 4 explains the implemented methodology for this study; Section 5 presents the findings of the study; the consecutive sections conclude the study through implications and limitations of the research, and end by providing the future research scopes.

2. Theoretical background

2.1. Leadership in social media and online research

The vast amount of literature in recent times regarding the appreciation of the nuance of leadership in the online context, especially in social media, draws attention to the understanding that scholars have developed about the impact that online leadership has on real-life events. Matthews et al., (2021)points out that since social media platforms are being used by leaders from all walks of life, such as business leaders, motivational/educational leaders, religious leaders, and rebellious leaders, amongst others, it becomes pertinent that the scholars of this domain extensively evaluate the contribution of this new and booming medium.

This also opens up the window of quantitative evaluation (Cortellazzo et al., 2019). This is because leaders like Elon Musk, Donald Trump, and Pope Francis, among other international leaders, have used and have been using social media platforms like Twitter, Instagram, and other social media platforms. Some use social media to instigate misapprehension about the business, political, or socio-technical climate (Heavey et al., 2020). Conversely, traditional leaders utilize these

platforms to adopt and proliferate traditional values in a modernized and pragmatic manner (Matthews et al., 2021). With open access to vast amounts of information, platforms like Twitter become valuable sources for studying social interactions, sentiment analysis, and content diffusion. Despite the potential for misinformation, these platforms also offer robust tools for capturing and categorizing the semantics of conversations, highlighting the dual nature of social media as both a disruptor and a unifier in the digital age (Martinčić-Ipšić et al., 2017).

There is a common understanding that advancement in any scholarly research is to figure out the movement of the evaluations being done in the research, look out for limitations and then appropriately address them (Cropanzano, 2009). But the academia has to have valid constructs for it to defend or argue against when trying to build on some of the understandings of a concept. This is important to understand because even though there has been a radical rise in social media (Khamis et al., 2017); it is still only a decade-and-a-half long (Van Dijck, 2013). Also, recent studies have shown that understanding the relationships between leaders and their followers are already on shaky grounds (Gottfredson et al., 2020), therefore using these half-a-century-old models of leaders and their impacts and relationships with their followers on social media or other online platforms would not provide comprehensively substantive results.

The issues identified by Gottfredson (et al., 2020) were so substantive that using the already existing models of leader and follower relationships and using them to quantify the social media relationships between the leaders and their followers sounded limitation-prone. But we believe that given the strengths of the technology that we have, we do not have to go back to the conceptual drawing board. Rather we could use the novel techniques of *Natural Language Processing* that have increasingly been used to validate or contradict the existing theories of varied domains (Birnbaum et al., 2017; Adamopoulos et al., 2018; Ramaswamy & DeClerck, 2018; Samuel & Zaïane, 2018; Bleidorn & Hopwood, 2019; Cheng & Tsai, 2019; Roy et al., 2020; Cartwright et al., 2021; Feeney & Porumbescu, 2021; Shankar & Tewari, 2021; Murthy & Lamkuche, 2022; Banks et al., 2022; Nguyen et al., 2022; Hodorog et al., 2022).

One important attribute that a study has found that influences online leadership is Self-Branding (Gilani et al., 2020), which has been used as a means to endorse a product by an organization through celebrities, but more recently, organizational and other domain leaders have been increasingly using self-branding as a tool to promote themselves and their ideas. This has grown so much that this type of leadership is now considered more authentic and trustworthy than the traditional forms of leaders and celebrities (Bladow, 2017). This has not been lost by businesses and organizations, as more of these institutions are actively engaging with these online leaders to reach their followers as their new and engaged target audience (De Veirman et al., 2017).

Social media, as a platform, provides an unobtrusive form of real-time primary data (Petrova et al., 2021) – and with this study, we have tried to demonstrate that just like the public interviews, statements, blogs, and reports, that have been traditionally used by scholars to examine leaderships, similarly, the social media usage by the leaders and their engagements with their followers can be used to build upon and develop a more comprehensive understanding of online or e-leadership (Avolio et al., 2000), especially in the social media realm. A comprehensive review study done by Matthews et al. (2021) indicated that the social media engagement of leaders is by far the most on Twitter – as it showed two-thirds of content on leadership generated on the micro-blogging giant, and hence was the target platform for our research. We also try and demonstrate that the more engaged the leader, the more would be his influential authority.

3. Data

There has been adequate literature that uses real-time social media data to understand the varied interactions between users (Chae, 2015; Daly et al., 2019; Kunst et al., 2019; Shankar & Tewari, 2021). This study also followed the framework through which manual classification of the tweets was done to classify and analyze these classified tweets. The initial extraction was done through the keywords like “leader”, “leadership”, “leadershipdevelopment” etc. after this, it was understood that the most prevalent hashtag would be #leadership or #leader, which were then finalized for this study.

This study was performed using Python (Version 3.9.8), with the help of the tweepy package of Python via Twitter, and Twitter API. The demo training data was extracted on 27th November 2021 (a small sample of 100 tweets), and the main data was extracted from 3rd December 2021 to 26th December 2021; which included 54,326 tweets containing leadership as their context and content. The collected tweets were public, and private tweets were excluded from the extracted list (Gokulakrishnan et al., 2012). Also, the confidentiality and privacy of the users were thoroughly maintained in this extraction and collection, as the personal, private, or confidential information regarding or for the users was completely excluded from the analysis of the study. Following Shankar and Tewari’s (2021) methodology, the study reinforces its understanding of leadership and how it impacts the relationships that develop between leaders and followers by drawing on methodologies endorsed by other prominent investigators (Chae, 2015; Yadav et al., 2022, 2023). This helped to reinforce the collection of 54,326 tweets over the course of a month, which is akin to the articles done in a similar domain. These investigations substantially enhance the methodological framework, increasing the study’s depth and relevance in associated areas. A summary of the dataset has been shown in Table 1.

3.1. Data preprocessing

Even though the tweets available from a user at any time are of three major types, i.e. Textual Tweets, Visual Tweets, or Audio Tweets, the tweets collected for this article were limited to textual tweets, as the analysis intended for this study would be focused to textual analysis of data. This process was done per the methods suggested by Shankar and Tewari (2021), by preprocessing and pretreating the data. This process led to the elimination of numbers, digits, URLs, contractions, negations, emojis and emoticons, non-ASCII characters (including UTF-8 Unicode), stop-words, extra spaces, slang, acronyms, colloquialisms, spellings, grammars, punctuations, texts to lowercases, mentions, and Stemming and Lemmatizing the words.

3.2. Algorithmic evaluation

With the help of data preprocessing, the contents of the tweets that have no substantially significant meanings are removed. After performing this process, the framework for Twitter text analysis was modified from the framework developed in the study done by Shankar and Tewari (2021) to analyze the Twitter dataset for the discourse about emotional intelligence, and it was then modified in this study to analyze the leadership interactions on Twitter. First, preliminary descriptive analytics was performed, followed by user and tweet statistics, to gather

Table 1
Summary of the dataset.

Dates of Data Collection	3rd December 2021–26th December 2021
Unique Users	16,053
Number of extracted Tweets	54,326
Number of hashtags in the tweets	2,71,732
Maximum number of Retweets on a tweet	59
Maximum number of favorites	546
Day with maximum tweets	Monday
Hour with maximum tweets	18:00
The date on which the maximum accounts were created of the users extracted	6th February 2013

information about the users and visualize the statistics of the data. This was followed by further clustering and classification of the words in these tweets through the topic modeling technique (Pak & Paroubek, 2010). This classification was then used to ascertain the users' general sentiments through sentiment analysis. This process categorized the tweets into three major polarities, i.e. positive, negative and neutral (Shankar & Tewari, 2021).

The categorization results of topic modeling and sentiment analysis were then aggregated to form the expression for principal component analysis. The aggregations achieved through the principal component analysis were then compared with the other aggregated outcomes to identify their semantic leanings, which were then used as the antecedents for the predictive analytics. The predictive analysis was done through the machine learning predictive model known as the Generalized Linear Model (GLM) (Friedman et al., 2010).

4. Results

4.1. Descriptive analytics

With the help of the Python package Gensim (Cao et al., 2009), we performed the descriptive analytics on the dataset, by using the understandings developed by Bruns and Burgess (2011) and the methodologies implemented in Shankar and Tewari (2021). This statistical technique is used as a building block for the analysis of data extracted or collected from the offline or online medium – for all businesses, leaders, followers as well as academics and researchers. This Gensim package of python was used to extract information about the tweets and their generators, and statistical techniques were then used on that information to analyze as well as visualize the components of the data.

4.1.1. User analytics and tweet statistics

Out of the 54,326 unique tweets extracted, the individual users accounted for 29.549 % (16,053 unique users), retweets were 29.68 % (16,126 unique retweets), and the mentions were 24.80 % (13,475 unique mentions) respectively. There were 2,71,732 hashtags used in these tweets, encompassing four important factors of leadership and especially the relationships between leaders and their followers in an online setting (Affect, Contribution, Loyalty, etc.). This means that there were, on average, 5.001 hashtags per tweet. Out of these total hashtags, 29,456 hashtags were unique, implying that there were significant repetitions of hashtags in the tweet texts.

On average, every individual user made 1.84 tweets, 3.36 retweets, and 0.233 mentions. These values helped us in evaluating user engagement on the platform. Active users were calculated using (tweets + retweets + mentions), whereas Visible users were calculated using (retweets + mentions) (Shankar & Tewari, 2021). Fig. 1 presents the graphical representation of the most active users against the most visible users. As expected, the most engaged users were the same as the most

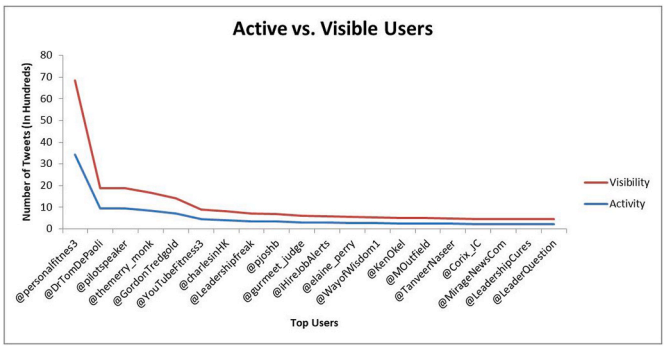


Fig. 1. Comparative representation of Active vs. Visible Users of the tweets extracted.

prevalently present users on the platform talking about leaders and leadership.

4.2. Topic modeling

For the process of topic modeling, the methodology of Latent Dirichlet Allocation (LDA) was implemented, using the process defined by Cao et al. (2009) on the Gensim package of Python. In this process, a corpus and dictionary of the most words was clustered, which acted as inputs to the designated LDA modeling. This was then followed by feeding the corpus into the pyLDAvis package (Mabey, 2020) for visualizing every significant topic and the distance of these topics from each other. Fig. 2 presents the Intertopic Distance Map of the distinct topics obtained through this corpus. It also shows that there were four distinct topics, with all four being significantly relevant clusters (Sievert & Shirley, 2014).

In order to get a better sense of these clusters, the primary hashtags allocated into these topics were identified, and collected. These collected hashtags from the relevant corpus provided us with a clear idea as to what these corpuses could possibly mean. Table 2 presents the 9 most relevant hashtags that were present in each corpus.

It was also found that out of 2,71,732 hashtags identified in the tweets, the most popular hashtags were *#leadership, #leader, #success, #management, #business, #motivation, #entrepreneur,* and *#culture*, amongst others. There were also 5.002 hashtags per tweets, with the top three hashtags accounting for 21.93 % of overall hashtags (59,570 times).

4.3. Sentiment Analysis

This data was subjected to sentiment analysis a package in Python known as Textblob (Zhang et al., 2018), which helped in classifying the emotions of these tweets into three polarities, i.e. Positive, Negative, and Neutral. This showed that the sentiment with which these tweets were generated were overwhelmingly positive, with a very small margin of tweets generating negative emotion. Fig. 3 presents the overall distribution of emotions of tweets and their text corpus.

Next, with the help of Textblob package of python, the words were classified into positive and negative sentiment, after which, the respective word clouds were created. Fig. 4 depicts the classification of words according to their emotions.

The word clouds show that *leadership*, *success*, *leader*, and *successful* are the most commonly occurring positive word, whereas *failure*, *sue*, *sad*, and *help* are the most commonly occurring negative word.

These topical sentiments were then standardized [−1, 0, 1] using the SentiStrength package of Python (Thelwall et al., 2011). They were then plotted on a sentiment polarity graph, which is shown in Fig. 5. This sentiment polarity graph showed a high positive skewed affinity of the dataset. A similar finding was found when the entire dataset was tested for sentiment analysis without the topics. Fig. 6 shows the non-topical sentiment analysis of the entire dataset. The apparent skew was leaning towards positive region of the graph, with 58.016 % of the tweets ranged between 0 and 1.

Even though sentiment analysis provides us with the users' views of the behavioral patterns of leaders and their social media engagements to their followers, it falls short in providing the factors that comprehensively lead to their favorable and unfavorable attitudes towards the leader's activities. In order to derive this causal relationship, a Principal Component Analysis followed by predictive modeling was done.

4.4. Principal Component Analysis

Topic modeling and sentiment analysis breaks the texts or sentences of texts into dictionary and corpus of words that provide us with a classification of the accompanying emotion as well as prevailing appropriate sentiment (Liu, 2022; Pang & Lee, 2008). Before conducting

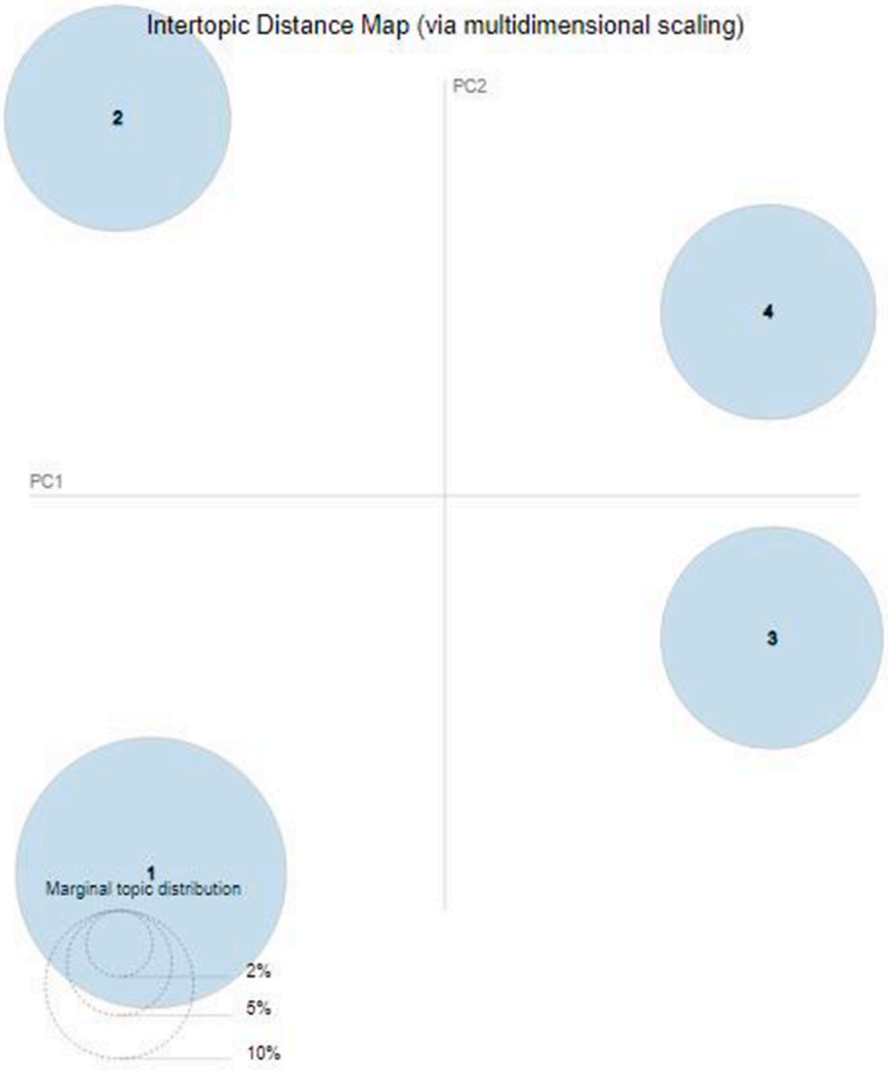


Fig. 2. Intertopic Distance Map of the corpus of words from the tweets to identify the distinct topics.

Table 2
Topic modeling and hashtags from the corpus of each distinct topics identified.

Perceived Leader Concepts	Frequency	Strategic Norms	Frequency	Intrapersonal Development	Frequency	Interpersonal Efforts	Frequency
#leadership	50102	#management	3800	#success	4605	#culture	1168
#leader	4863	#business	3610	#motivation	2261	#job	607
#leadershipdevelopment	2202	#entrepreneur	1769	#innovation	1525	#team	550
#leaders	1640	#soar2success	1285	#inspiration	1289	#learning	528
#inspiration	1289	#strategy	1069	#personalgrowth	1063	#training	528
#thoughtleadership	565	#ceo	1050	#mindset	958	#change	518
#leadershipmatters	433	#goals	795	#change	518	#communication	512
#excellence	356	#career	602	#emotionalintelligence	313	#partnership	498
#leadershipcoaching	265	#itstrategy	475	#attitude	345	#teamwork	482

any PCA test, the two tests, i.e. Kaiser-Meyer-Olkin (KMO) test and Bartlett's test for sphericity were conducted to check the suitability of the data for multivariate analysis (Hair et al., 2006).

The KMO test is used to measure the multicollinearity amongst the dataset, and the Bartlett's test of sphericity is used to identify the correlation amongst the variables. A high value of KMO test (maximum is 1) and the p-value of Bartlett's test warrant the use of PCA for analysis (Hair et al., 2006). In our study, python package Scikit-learn (Paper & Paper, 2020, pp. 1–35) was used to measure the values of KMO test and Bartlett's test. The KMO test value for our dataset was found to be 0.961, with Bartlett's test of sphericity's p-value being 0.000, indicating that a PCA test is warranted for our particular dataset.

The PCA was used to find a set of Principal Components (PC) with a loss of minimal information. The process used in this analysis aggregated the probabilities of the sentiment polarities of each word in the topics, providing weights to each word. This was then compared with the original topic modeling algorithm for validation, which provided us with four distinct clusters, giving us four principal components. The weights were provided based on their positivity, and the outliers were removed. Table 3 presents the top 10 words in each of the components

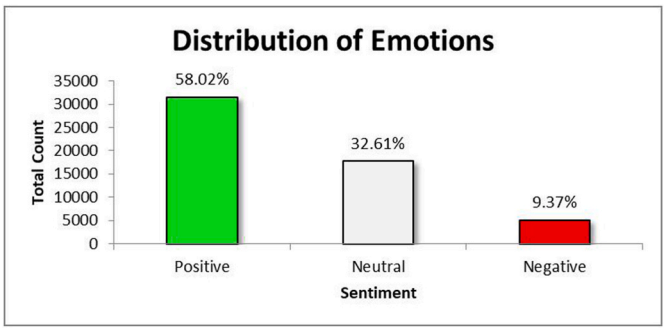


Fig. 3. Percentage distribution of sentiment of emotions in tweet texts.

based on their weights.

Stemming of the words of the texts let us only find the root words of the texts, which is also presented in Table 3. Since the study is trying to evaluate the user's behavior regarding the discussions about the leader's behavior on the online platforms and to their followers, the first component was selected for further analysis, as it presented the most relevant words related to leaders' behavior online as well as towards their followers.

4.5. Predictive modeling

The third component was used to predict the users' behavior based on the public engagement done by the authors (tweet/tweet thread creators) regarding a leader's discourse on social media. A multivariate normality was measured through the Shapiro-Wilk test by using the python package SciPy (Virtanen et al., 2020), which showed that the data of the components were non-normalized in nature (p-value = 4.179e-5), which in turn, prohibited the use of any linear regression models for the predictive analysis. Instead, the study used a Generalized Linear Model (GLM) (McCullagh & Nelder, 2000) in order to establish and predict the relationship between the driving factors that affect a user's tweeting behavior on the Twitter platform.

The users' behavior predictors formed due to the weighted terms contained in PC1 are (1) polarity – frequency of positive and negative words, (2) count – number of tweets by a user, (3) qtweets – number of quote tweets by a user, (4) favorites – a particular set of topics or subjects a user is most interested in, (5) followers – number of followers a user has, and (6) countfreq – total number of tweets made by a user. The GLM allows the predictive linear regression by relating the response value through a link function by using variance of each calculation of its expected value (McCullagh & Nelder, 2000). The GLM analysis was performed using python's package Statsmodels (Seabold & Perktold, 2010). The findings of the GLM statistics are presented in Table 4.

As can be seen from the GLM table, the model is impartial and is significantly just estimated. The residual dependence of the values was almost near to zero (with reduced degrees of freedom), which also showed that the model was statistically significant (Casals et al., 2014), and the goodness of fit was also measured using Pearson's Chi-Squared test and Akaike's Information Criterion (AIC); with Chi-Squared value being 2.19e+04 (or 21900) and the ΔAIC being 4.5 between the final model and a previous model, all showing that the model is satisfactory. The Pearson Residuals did not have a patterned outcome, suggesting that the data was randomized and the outcome has not favored any of the input variables, also telling that the model is appropriate.

The validation of this model was also done through calculating the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The MAE of this model was 0.0387 and the RMSE was estimated to be 0.1967, which suggested that since these values are very small (Chai & Draxler, 2014), stating that this predictive model has performed well.

5. Discussion

This study was conducted using 54,326 tweets analyzing the

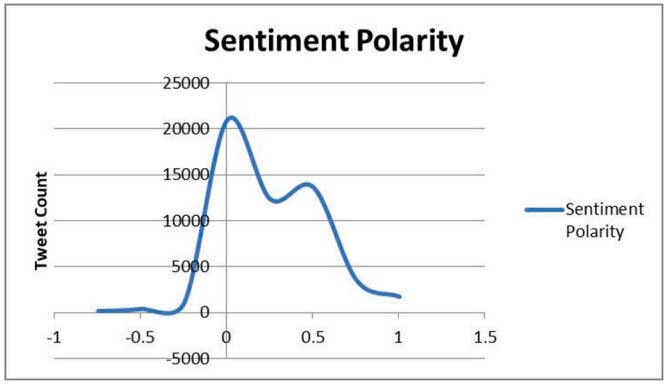


Fig. 5. Standardized distribution of the sentiment analysis of the tweet text.

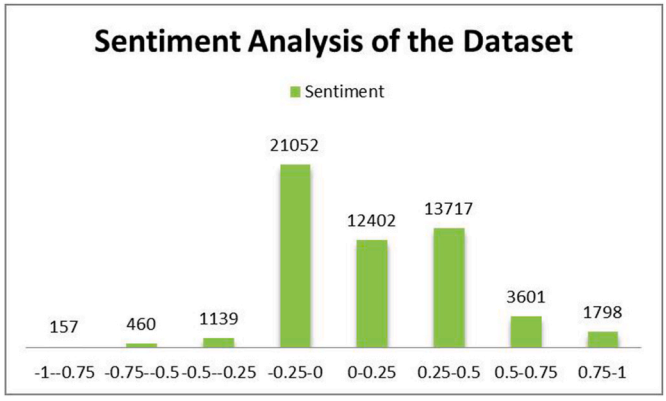


Fig. 6. Sentiment Analysis of non-topical level of entire tweet dataset.



Fig. 4. Word cloud of Positive as well as Negative words of the extracted tweets.

Table 3
Top 10 words in the Principal Components extracted from the tweet texts.

Term	PC1	Term	PC2	Term	PC3	Term	PC4
leader	0.083	business	0.040	success	0.060	today	0.027
share	0.028	team	0.031	people	0.041	go	0.018
management	0.025	work	0.027	new	0.019	read	0.017
learn	0.024	great	0.022	culture	0.017	skill	0.016
follow	0.024	change	0.021	help	0.016	think	0.014
need	0.018	lead	0.021	thing	0.016	create	0.013
time	0.016	good	0.021	keep	0.014	strategy	0.013
life	0.015	podcast	0.015	well	0.014	technology	0.011
innovation	0.014	see	0.013	look	0.011	disability	0.011
want	0.014	become	0.010	world	0.010	relationship	0.010
Variance (Explained)	0.006		0.013		0.008		0.007
Standard Deviation (Explained)	0.077		0.114		0.089		0.084

Table 4
Regression statistics using Generalized Linear Model on the PCA identified.

Generalized Linear Model Regression Results						
Coefficients	Estimate	Std. error	t value	P>|t|	Lower Bound	Upper Bound
Constant	*−3.427*	*0.044*	*−78.651*	*0.000***	*−3.513*	*−3.342*
polarity	1.331	0.023	57.409	0.000***	1.286	1.377
count	−7.23e-07	8.81e-06	−0.082	0.935	−1.80e-05	1.65e-05
qtweets	−5.55e-08	8.84e-08	−0.628	0.53	−2.29e-07	1.18e-07
favorites	0.008	0.005	1.520	0.128	−0.002	0.019
followers	−0.019	0.014	−1.393	0.163	−0.047	0.008
countfreq	−0.038	0.123	−0.315	0.753	−0.280	0.203
*Significance: *** = statistically significant*						
Pearson Chi-Square	*2.19e+04*				*Family*	*Binomial*
Degrees of freedom of Model	*6*				*Method*	*IRLS*
Degrees of freedom of Residuals	*54319*				*Data*	*Leadership*
Number of Iterations	*6*				*Link Function*	*logit*

characteristics of leaders and their leaderships on social media platforms, by identifying the factors that drive these leader's and other users' tweeting behavior frameworks on social media. In the literature collection and secondary analysis, it was found that even though there have been some prevailing studies that evaluate similar relationships to this study, the basic construed premise of online leadership engagement and their impacts on other users and followers' behavior is sporadically studied (Bladow, 2017; Gottfredson et al., 2020). This study was thus carried out to start a discourse about the online leadership behavior and its overall impact on the online discourse.

The proportion at which the social media users engaged in the conversations about leaders and leadership were similar to previous studies (Matthews et al., 2022; Johnson et al., 2015), as through data collection, it was found that discussions were fairly steady across the timeframe, but an interesting findings was the overwhelming use of hashtags (see Table 1). This was unexpected given the domain of query, but was still consistent with other findings (Shankar & Tewari, 2021). An average of 5.001 hashtags were used per capita tweets, and an expected high number of #leadership were found to be used in these tweets.

Nearly all of the tweets were originally created tweets, as there were marginal number of retweets and quote tweets as compared to originally generated tweets. A surprising number of leading handles were of fitness and health, closely followed by business, subject matter experts, and opinion leaders (see Fig. 1). The most active users were the most visible users, which is consistent with previous studies (Bovet & Makse, 2019). Top 4 % of the users accounted for 54 % of the tweets, which is also consistent with the previous studies (Shankar & Tewari, 2021; Zafar et al., 2021).

The sentiment analysis found that the Twitter users hold an overall positive sentiment towards a leader and leadership framework and have a high concern for the decentralized nature of a leader in social media sphere, because of the amount of freedom being accosted to everyone on these platforms, including leaders. The negative emotion accounted for a marginal 9.87 % of the overall emotion, which indicated that users had minimal negative attitude towards a leadership framework (see Fig. 3). An interesting finding was a huge amount of neutral sentiment, which accounted for 32.61 %, contrary to previous findings (Bonilla & Rosa, 2015; Bovet & Makse, 2019), which can be accredited to the nature of the conversations around leadership, which is often generic, untargeted, and even preachy in nature.

Good topic modeling outcome was achieved with the help of LDA model, which provided us with four distinct topics around which the sentiment corpus and predictive analytics was developed. These four topics were then inserted into the predictive algorithms which provided us; again, four prominent principal components (see Fig. 2 and Table 3). The topics identified were perceived leader concepts, strategic norms, intrapersonal development, and interpersonal efforts. This builds on our understanding as previously challenged by Gottfredson (et al., 2020), which argues that there is a need to significantly study the leader follower relationship constructs through media other than qualitative, in exploratory manner. This finding also provides a new avenue through machine learning, which has been lacking in understanding leader and follower relationship, especially in online platforms (Cortellazzo et al., 2019; Gilani et al., 2020).

For the predictive analytics, principal components were obtained. The high value of KMO and Bartlett's test allowed us to measure the PCs of the dataset, which gave us four significant components. Out of the four components, the first component provided richer image based on their component weights and their component variance and deviation (see Table 3). The lower PC values were talking about the abilities that a leader must foster as leadership qualities, and the higher PC values were talking about the environmental factors that lead to a leader identification and development. The causation of these components' attitude formation was measured through the generalized linear model, which

stated that the model has predominantly statistically significantly negative effects. This negative nature has been concurred in other studies (Guo & Zhou, 2016; Banks et al., 2022; Scandura & Meuser, 2022) and the factors driving the effects were majorly aversive.

It was found that intention, psychological safety, and trust were amongst the leading factors driving the negative sentiment and attitude towards the discussions regarding leadership and one's leader, which was apparent in the corpus of words formulated during corpus (see Fig. 4). In the predictive analytics, it was found that other than sentiment polarity, all the other external constraining factors are statistically insignificant, which means that only sentiment polarity builds a user's behavior towards a leader and overall leadership discussion on Twitter, whereas other factors do not contribute to building one's behavior towards leadership discourse. Sentiment polarity of the textual content (i.e. negative or positive emotion towards the tweet texts) has a statistically significant influence on the overall leader and leadership discussion on Twitter. This would mean that a user does not overly rely on other information that helps in circulation of Tweets on Twitter when it comes to a leader or leadership discussion, rather they rely on the positivity or negativity of the tweets that are being circulated, written or read by people on Twitter. This means that users are more reliant on non-authoritative handles and platforms regarding these discussions, which can also be cross-referenced through the calculation of active users (see Fig. 1). This provides an important insight to the leaders and organizations that can help create a more informal environment around the leader for a more engaged online audience (Agarwal & Toshniwal, 2020; Shankar & Tewari, 2021).

6. Study implications

A substantial increase in the use of Twitter in academic and corporate world can be seen in recent years (Cripps et al., 2020; Culnan et al., 2010); and the findings of this study further reinforce the understanding that professional leaders, as well as leaders of other domains, must use Twitter as a platform for discussing leaders and leaderships. This is important for leaders in public space, including business organizations, because the environment, both organizational and non-organizational, are ever-dynamic and ever-changing, and the leaders have to be adaptive and proactive in building their styles and portfolios that continuously and efficiently change and adapt to the changing environment.

The findings of this study are in fact, inter-disciplinary: domains of business, psychology, leadership, media and communication, politics, behavior and information technology, amongst others can build on these foundations. This area is not in any manner less evaluated, but it has either been studied in a very broad sense, or in pretty small pockets of dimensions. In terms of understanding the concepts of relationship between a leader and their follower, studies have even questioned the validity of the constructs (Gottfredson et al., 2020; Scandura & Meuser, 2022). The study, thus, extends the findings of leadership and the understandings of leader follower relationship in online media, by providing a working and viable construct through the textual corpus of topic modeling and PCA.

This study also provides a linkage between the causative factors that drive a sentiment inclination, tweeting pattern and online behavior of users towards a leader and leadership. This study also provides a schema of identifying prominent users that can be identified as leaders on Twitter. This is crucial for users twofold: one, it provides users with an insight as to who can be identified as a leader, driving the leadership discourse; and two, for leaders and marketing as well as public relations organizations, as it provides them with a more comprehensive understandings of using these handles for advertisement, promotion, as well as engagements regarding their ideas, products, and other marketable entities.

The current findings demonstrate that a user's predefined values and beliefs, pooled with tweet polarity, might provide a comprehensive understanding about the origins of a leader's behavior on social media. Since polarity also has a strong foundation on the constructs of relationship between a leader and their followers, this, combined with high engagement between users and leaders, provide a substantial insight into measuring a user's predisposition towards the use of Twitter as a viable medium for online leadership discussions.

7. Limitations and future research

The increasing interests of professionals, organizations, and academicians in understanding leadership in public domain have warranted an important outcome: Twitter is an important source of information for leaders in understanding their business, ideological, pragmatic, and decisional stances. This study has tried to build on this outcome. But as with any research, this study also has limitations. One limitation is the data itself. The data has been collected in a very short period of time, and thus, an increased timeframe of data collection for a similar study would increase our dataset for a more comprehensive analysis. The dataset is also rather modest, as Twitter is capable of providing a larger corpus of data for similar analysis.

Another limitation is the unit platform, as future studies can inculcate other social media and info-media platforms for a more comprehensive understanding of our constructs. The method of data collection can also be worked with, as more detailed and specific preferential keywords can be used to paint a better picture of the findings. The future studies may also study the impact of engagements and connectedness of specific users, such as already established leaders in a public or private domain, opinion leaders, and other leading entities, which would help in understanding the complex nature of Twitter's interactions by the users and information prosumers.

8. Conclusion

As the world comes out of an unprecedented time, understanding our leaders and how they operate, especially in an online environment, has become increasingly necessary. The factors that drive a leader's engagement, connectedness, relationships, and behaviors have to be studied in varied contexts. It does not help the fact that all of these factors have an overlapped relationship with themselves traditionally, but in an online context, these differences have become particularly defined. And as Gottfredson (et al., 2020) stated that the evaluation of leader relationships with their followers has been fundamentally theoretically flawed, and it is imperative that the future evaluators use the resources at their disposal to re-study these phenomenon. This study tried to provide the evidentiary argument to this query that we do not need to go back to a proverbial drawing board to understand the mechanics of leadership and their follower relationships, rather, use the advanced tools available to us to build on the already developed knowledge in this area.

This study tried to evaluate the use of Twitter as a tool for public discourse and information sharing, and a means to generate positive intentions in its users regarding the leaders and leadership. The results were analyzed through a comprehensive NLP technique for textual analysis, topic modeling, and predictive analytics. These analyses presented us with four distinct topics, which were further validated through four components, which showed that even though the Twitter users are negatively inclined towards trusting and engaging with a leader or discussions about leadership, a change in textual polarity might sway their opinions and emotions regarding a discussion issue. It also clearly showed that just by simply increasing the data corpus is not going to change this outcome; rather increasing the intensity of textual polarity would bring in a significant change in these findings. Further studies can include other forms of data, as well as other analytic techniques, including deep learning and neural network techniques to better enhance the patterns in the data. This study, thus, opens up more avenues for both academicians as well as professionals to further increase their engagements and relationships that would help in improving their

overall outcomes.

CRediT authorship contribution statement

Shardul Shankar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vijayshri Tewari:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: NA.

Data availability

The data and associated code for this study are available on the corresponding author's GitHub profile. Access the repository at: https://github.com/theshane007/twitterleadership.

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